Occlusion-Resistant Object Detection for Dual-View Airborne Platforms

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***Abstract*—Infrared object detection in airborne surveillance faces significant challenges, particularly when targets are occluded and viewpoints are sparse and dynamic. Existing multiview methods typically rely on dense camera arrays or precise static calibration, rendering them ill-suited for ad hoc dual-uncrewed aerial vehicle (UAV) deployments characterized by asynchronous capture and sensor noise. To bridge this gap, we propose a dual-platform collaborative occlusion-resistant framework tailored for resource-constrained airborne platforms. This system-level architecture integrates three synergistic components to address these constraints: 1) a hybrid geometric–semantic registration (HGSR) scheme that establishes a robust spatial foundation by combining coarse physical projection with fine-grained semantic correction to align asynchronous views; 2) a multiscale enhanced dual-domain strip attention module (MS-DSAM) that extracts occlusion-invariant features by jointly modeling spatial strip patterns and frequency-domain spectral consistency; and 3) a dual-view localization loss (DLL), formulated as a weakly supervised regularization mechanism, which enforces cross-view semantic and geometric consistency without requiring dense supervision. Extensive experiments on the self-constructed DualView-Occlusion dataset and standard benchmarks (Wildtrack and MultiviewX) demonstrate that our method achieves 58.9% mAP50 on small targets under moderate occlusion (Dataset A) using only two mobile views, outperforming state-of-the-art methods that rely on three or more fixed views. The system operates at 23.31 FPS, meeting real-time requirements for UAV edge deployment.**

***Index Terms*—Dual-view fusion, localization loss, occlusion-resistant detection, Siamese network, spatial registration.**

I. Introduction

INFRARED target recognition serves as a foundational capability for airborne surveillance, autonomous navigation, and cooperative robotics. While single-view systems are widely deployed due to their simplicity, their fixed perspective renders them highly vulnerable to occlusion, leading to severe information loss and feature ambiguity—particularly in textureless thermal imagery. Multi-view paradigms, especially those leveraging dual airborne platforms, offer a promising alternative by fusing complementary observations to recover occluded semantics and enhance geometric reasoning (as conceptually compared in Fig. 1).

The limitations of single-view detection under occlusion are empirically evident. When evaluated with state-of-the-art detectors (e.g., YOLO series) on the heavily occluded Wildtrack dataset, detection accuracy drops to 85.5%. On our more challenging *DualView-Occlusion* dataset—designed to simulate realistic UAV scenarios with dynamic motion and partial visibility—the accuracy further declines to 59%, accompanied by frequent missed detections and false positives. These results underscore the inadequacy of monocular approaches in occlusion-prone airborne environments (see Fig. 2).

In response, researchers have explored multiview methodologies [1], [2], [3]. However, prevailing approaches—particularly those developed for occluded pedestrian detection in datasets such as Wildtrack or MultiviewX—assume a static, densely distributed camera array with 3–7 synchronized and pre-calibrated viewpoints. Such configurations are fundamentally incompatible with resource-constrained uncrewed aerial vehicle (UAV) operations, where only *two asynchronous, dynamically moving platforms* are typically available. As we empirically demonstrate, naively adapting these frameworks to sparse UAV settings results in severe performance degradation due to their reliance on dense geometric coverage and precise extrinsic calibration.

This mismatch stems from three critical limitations in existing multiview object detection paradigms when deployed in airborne infrared scenarios.

1) *Sparse and Dynamic Viewpoints:* Ground-based surveillance systems benefit from fixed, overlapping fields of view that enable robust triangulation and feature matching. In contrast, UAV formations often operate under strict payload and communication constraints, limiting observations to dual, nonoverlapping, and time-asynchronous views. Traditional architectures exhibit sharp performance drops under such sparsity, as they lack mechanisms to compensate for missing perspectives.
2) *Sensor Uncertainty and Misalignment:* High-precision extrinsic calibration is routinely achievable for terrestrial cameras but remains impractical for small-scale UAVs subject to wind gusts, rapid maneuvers, and GPS/IMU drift. Consequently, geometric projection based on noisy pose estimates introduces significant misalignment during cross-view feature fusion—a problem that purely spatial or attention-based alignment schemes fail to resolve.
3) *Occlusion in Textureless Infrared Imagery:* Thermal imaging, while advantageous for all-weather opera-

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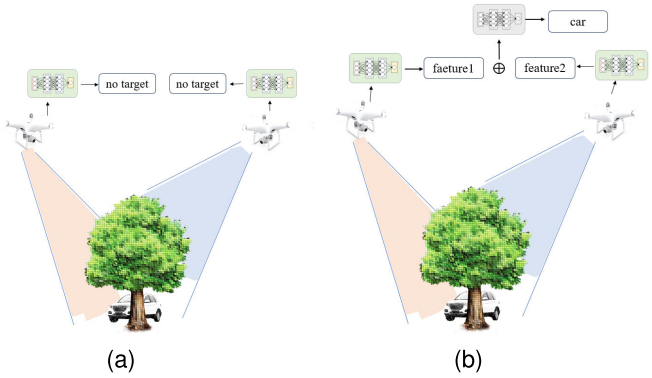


Fig. 1. Comparison diagram of (a) single-view and (b) dual-view framework.

tion, suffers from low texture and absence of color cues. Under occlusion, conventional spatial attention modules cannot reliably disambiguate overlapping targets, leading to missed detections or identity confusion—especially when complementary views are not effectively leveraged.

To address these limitations, this article proposes a dual-view collaborative occlusion-resistant object detection network (DVC-ORN), specifically designed for airborne infrared surveillance under the operational constraints of UAV swarms: extreme viewpoint sparsity, asynchronous sensing, and severe occlusion in low-texture modalities. The framework introduces three tightly integrated innovations: 1) a multiscale enhanced dual-domain strip attention module (MS-DSAM) that jointly models spatial structure and frequency-domain invariance to preserve discriminative cues under partial occlusion; 2) a hybrid geometric–semantic registration (HGSR) module that achieves precise cross-view alignment by fusing noisy but informative UAV pose data (IMU/GPS) with semantic feature correspondence—eliminating reliance on perfect calibration while tolerating moderate sensor drift; and 3) a dual-view localization loss (DLL) that enforces geometric and semantic consistency between views through a combination of cosine similarity and SmoothL$_1$ regression, thereby regularizing feature learning even when one view is heavily occluded. Collectively, these components enable robust detection in real-world dual-UAV scenarios where conventional multiview methods—designed for dense, static, or well-calibrated setups—fail to generalize.

The main contributions of this article are summarized as follows.

1) We propose a dual-branch Siamese network integrated with MS-DSAM—a pioneering module that uniquely integrates spatial strip patterns and frequency-domain spectral consistency from only two mobile viewpoints, significantly enhancing feature robustness under occlusion.
2) We design DLL, a specialized loss function that combines cosine similarity for semantic consistency with SmoothL$_1$ regression on UTM-projected bounding boxes, enabling cross-view alignment without fixed camera calibration.
3) We develop a lightweight spatial registration pipeline (HGSR) that leverages real-time IMU/GPS data to align asynchronous dual-view features into a global coordinate system, supporting dynamic UAV flight.
4) We construct and release the DualView-Occlusion benchmark dataset—a comprehensive resource providing synchronized infrared–visible pairs and precise UAV pose metadata, which fills the gap in existing datasets for sparse-view airborne surveillance.

The remainder of this article is structured as follows. Section II reviews the state of the art in single-view and multiview object detection. Section III elaborates on the proposed methodology. Section IV presents the experimental results and a comprehensive analysis. Finally, Section V concludes this article and outlines directions for future work.

II. Related Work

Multiview data are ubiquitous in airborne surveillance. Data from disparate viewpoints inherently possess distinct representations and statistical properties, offering complementary information that single-view methods fail to capture. The core challenge in multiview target recognition is twofold: efficiently fusing these complementary features while mitigating interference from severe occlusion, sensor noise, and viewpoint variations. This section reviews the evolution of these techniques, ranging from single-view baselines to advanced multiview paradigms.

A. Single-View Detection and Limitations

Object detection has historically focused on single-view scenarios, establishing a mature technical framework categorized by architecture and optimization strategies. Two-stage detectors, such as the R-CNN family [4], [5], achieve high precision via region proposal refinement, while one-stage detectors like FCOS [6] and CenterNet [7] prioritize speed through end-to-end prediction. To address class imbalance during training, Lin et al. [8] introduced focal loss, and Rezatofighi et al. [9] proposed the G-IoU loss to optimize bounding box regression.

In the infrared domain, specific challenges such as low texture and background clutter have been addressed through domain adaptation and local contrast learning [10], [11]. To counter occlusion, researchers have explored part-based models [12], [13] that decompose targets into visible components, and specialized loss functions like repulsion loss [14] to penalize bounding box overlap in crowds.

Despite these advancements, single-view methods are fundamentally limited by the "line-of-sight" constraint. When critical features are obscured or targets are truncated, monocular approaches lack the geometric priors needed for feature completion, leading to inevitable missed detections. This necessitates the transition to multiview frameworks, particularly for complex airborne environments.

B. Multiview Detection and BEV Paradigms

Multiview detection has evolved from handcrafted geometric alignment to deep feature fusion. Early approaches relied

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on precise calibration to align multiaspect images, using techniques like rotational pseudo-color synthesis [15] or mutual information maximization [16]. While effective in controlled settings, these methods are brittle under the dynamic pose variations typical of UAVs.

The advent of deep learning introduced unified bird's-eye-view (BEV) representations. The lift-splat-shoot (LSS) framework [17] established a paradigm for encoding images from arbitrary rigs into a 3-D BEV space without explicit point clouds. Subsequent works like MVDet [18] and MVDetr [19] adapted this for pedestrian localization, achieving state-of-the-art performance on ground-based benchmarks. Recent advances further refine this pipeline. SHOT [20] employs stacked homography transformations to mitigate perspective distortion, while 3DROM [21] introduces 3-D occlusion augmentation to improve robustness. Additionally, Zhang et al. [22] proposed a Mahalanobis distance-based optimal transport loss to refine localization in crowded scenes.

However, deploying these BEV and Transformer-based architectures on UAVs faces two critical bottlenecks.

1) *Computational Prohibitiveness:* Transformer-based cross-view attention exhibits a computational complexity of $\mathcal{O}(N^2)$ with respect to sequence length N. Given the high-resolution requirements of infrared aerial imagery, full-scale MV-DETR models drastically exceed the memory and FLOP limits of edge devices, rendering them too heavy for real-time onboard processing.
2) *Sensitivity to Calibration:* BEV reconstruction fundamentally depends on the inverse perspective mapping (IPM) derived from accurate extrinsic matrices. In dynamic UAV scenarios, inherent GPS/IMU telemetry noise introduces nonlinear errors. Even minor attitude deviations (e.g., pitch or roll drift $>2^\circ$) propagate exponentially into the projection homography, causing severe pixel-level spatial misalignment within the discrete BEV grid. This breaks the geometric correspondence necessary for dense feature matching.

To address these limitations, our work bypasses explicit 3-D reconstruction, opting for a lightweight Siamese architecture that performs implicit feature alignment.

C. Attention Mechanisms and Robust Fusion Strategies

Parallel to BEV methods, research into lightweight fusion and attention mechanisms offers a viable path for resource-constrained platforms. Biologically inspired paradigms have demonstrated that mimicking human visual attention can achieve high perceptual fidelity with minimal overhead. For example, Khan et al. [24] proposed an attention-enhanced machine instinctive vision framework that significantly improves saliency detection efficiency. Similarly, multiscale trapezoidal attention networks [23] have been developed to capture features at varying granularities, which is crucial for detecting targets of different scales in aerial imagery.

Addressing the challenge of fusion under uncertainty, recent works have explored robust integration strategies. The concept of bilateral feature fusion with hexagonal attention [25] highlights the importance of adaptive weighting when combining information from uncertain environments. Drawing from these insights, we argue that for anisotropic targets (e.g., vehicles and pedestrians) in infrared imagery, standard square pooling kernels are suboptimal. Instead, we adopt the strip pooling concept introduced by Hou et al. [26], which captures long-range contextual dependencies along narrow directions.

Inspired by these foundational works, our proposed MS-DSAM extends strip attention into both spatial and frequency domains. This design effectively captures the "long and narrow" characteristics of occluded targets while leveraging frequency-domain invariants to maintain robustness against sensor noise and viewpoint shifts.

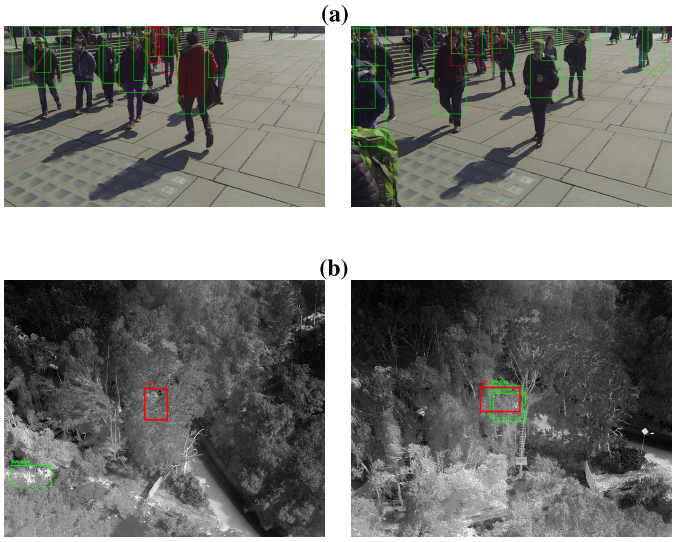


Fig. 2. Visualization of single-view detection failures under target occlusion. Green boxes indicate correct detections, and red regions indicate missed detections. (a) Single view on Wildtrack dataset. (b) Single view on DualView-Occlusion dataset.

D. Gap Analysis

Despite significant progress, existing multiview paradigms leave a critical gap in airborne infrared surveillance due to three fundamental misalignments with UAV operational constraints.

1) *Rigidity Versus Dynamism:* Current registration methods (e.g., homography-based or BEV-based) assume either static sensor poses or high-precision synchronization. They lack a *hybrid mechanism* to tolerate the nonlinear spatiotemporal jitter inherent in asynchronous dual-UAV flight, leading to feature misalignment in edge conditions.
2) *Spatial Dependency Versus Occlusion:* Prevailing fusion strategies rely heavily on spatial correlation arrays (e.g., nonlocal blocks or standard self-attention). When physical occlusions fragment target topologies, continuous local receptive fields are abruptly truncated, leading to a catastrophic drop in feature activation. These methods overlook frequency-domain invariants—specifically, the translation-independent, low-frequency spectral energy

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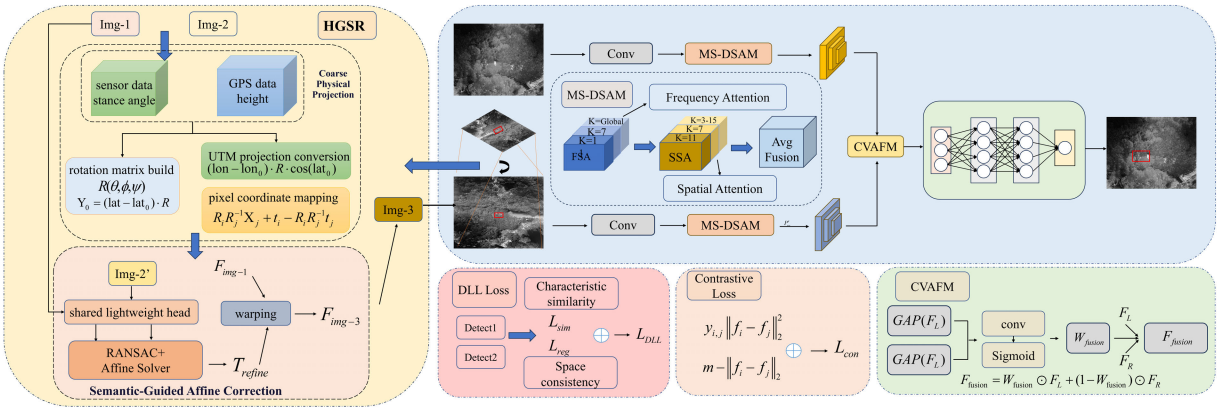


Fig. 3. Overview of the proposed dual-platform collaborative occlusion-resistant object detection network. The framework operates on asynchronous dual-view infrared inputs and consists of four key stages: 1) HGSR for robust alignment under sensor noise; 2) dual-domain feature extraction using a Siamese ResNet-50 backbone enhanced with MS-DSAM to capture occlusion-invariant features; 3) cross-view adaptive feature interaction module (CVAFM) for dynamic fusion; and 4) a collaborative loss suite (L_{DLL} and L_{con}) to enforce geometric and semantic consistency. This design enables accurate detection under severe occlusion using only two uncalibrated, mobile viewpoints.

characteristic of thermal targets—which can bridge fragmented spatial features and maintain representation integrity despite corrupted spatial cues.

3) *Supervision Cost versus Scalability:* Most state-of-the-art methods (e.g., MV-DETR) depend on dense pixel-level supervision or precise calibration for training. They fail to exploit the weak supervisory signals inherent in the physical geometry of dual-platform systems, restricting their deployability in uncalibrated, label-scarce environments.

To bridge these gaps, our work bypasses the requirement for perfect calibration and dense supervision, proposing a system-level solution that integrates geometric constraints with semantic learning.

III. Methodology

To address the challenges of prohibitive annotation costs and limited feature robustness in sparse-view UAV surveillance, we propose a dual-platform collaborative occlusion-resistant framework. As illustrated in Fig. 3, this architecture is designed not as a collection of isolated modules, but as a unified system driven by the logical coupling of geometric alignment and semantic representation. The framework operates through three tightly integrated stages.

1) *Geometric Foundation:* An HGSR module first establishes a shared coordinate system from asynchronous inputs, providing the precise pixel-level alignment that serves as the prerequisite for effective cross-view interaction.
2) *Representation Enhancement:* Building upon this spatial alignment, the Siamese network integrates an MS-DSAM to extract feature representations that remain discriminative even when structurally corrupted by occlusion.
3) *Weakly Supervised Regularization:* Finally, a DLL exploits the established geometric correspondence to enforce semantic consistency, acting as a weakly supervised mechanism that uses physical alignment as a supervisory signal to guide feature learning.

A. Dual-View Spatial Registration With Hybrid Geometric–Semantic Refinement

Effective fusion in sparse dual-view scenarios depends critically on precise pixel-level alignment. However, in airborne infrared surveillance, we identify two distinct *failure modes* that render conventional single-domain registration methods ineffective.

1) *Geometric Failure Mode:* Methods relying solely on geometric projection (using GPS/IMU) suffer from nonlinear misalignments due to sensor drift and temporal asynchrony. As shown in our experiments, even minor attitude errors (e.g., yaw drift >2°) can lead to projection disparities exceeding 30 pixels, breaking the correspondence needed for feature fusion.
2) *Semantic Failure Mode:* Traditional visual registration (e.g., SIFT, ORB, or optical flow) fails in infrared imagery due to the lack of textural details and the significant perspective changes between widely separated UAVs. In these textureless environments, visual descriptors cannot find sufficient stable keypoints to compute a reliable homography matrix.

To resolve this dilemma, where neither geometric nor semantic methods can operate reliably in isolation, we propose an HGSR framework. This module operates on a "coarse-to-fine" principle. It uses noisy physical data to constrain the search space (overcoming the semantic failure) and uses sparse semantic semantics to correct sensor drift (overcoming the geometric failure).

1) Stage 1 [Coarse Physical Projection (Geometric Initialization)]: We first map pixels from the source view to a unified world coordinate system using UAV telemetry. This step provides a global geometric prior, ensuring that the target search range is confined to a physically plausible region, regardless of image texture. Let $P_{uv} = [u, v, 1]^T$ be a pixel in the image plane. Its corresponding world coordinate P_{world} is derived via

$$P_{world} = \mathbf{R}(\theta, \phi, \psi)^{-1} \cdot \mathbf{K}^{-1} \cdot P_{uv} \cdot h + T_{GPS} \tag{1}$$

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where $\mathbf{R}$ denotes the rotation matrix derived from the IMU (pitch ψ, yaw θ, and roll ϕ), $\mathbf{K}$ is the camera intrinsic matrix, h is the relative altitude, and T_{GPS} represents the UTM-converted translation vector.

While this projection is inherently noisy due to sensor inaccuracy, it effectively neutralizes large-scale viewpoint disparities, transforming a "global search" problem into a "local correction" problem.

2) Stage 2 [Semantic-Guided Affine Correction (Residual Compensation)]: To eliminate the residual geometric distortions left by Stage 1 (e.g., perspective skews caused by microfluctuations in pitch angle), we employ a lightweight semantic refinement. Unlike traditional dense feature matching, we extract high-level semantic keypoints (specifically, object centroids detected by a shared lightweight head), which remain stable even in low-texture infrared images.

Let $\{k_i\}_{src}$ and $\{k_i\}_{dst}$ denote matched semantic keypoints between the projected source view and the target view. We compute an optimal affine transformation matrix $\mathbf{T}_{refine}$ via RANSAC to minimize the alignment error

$$\mathbf{T}_{refine} = \arg\min_{\mathbf{T}} \sum_i \left\| \mathbf{T} \cdot k_i^{src} - k_i^{dst} \right\|_2^2. \tag{2}$$

This affine transformation $\mathbf{T}_{refine}$ serves as a residual correction term. It specifically compensates for planar homography distortions induced by attitude noise, which pure geometric projection cannot predict.

The final registered feature map is obtained by warping the source features using $\mathbf{T}_{refine}$. By integrating these two stages, HGSR achieves robust alignment (<5 pixels error) in scenarios where GPS errors (up to 5 m) would break geometric methods and textureless backgrounds would break visual methods.

B. Dual-Domain Feature Extraction and Fusion

We employ a weight-sharing Siamese ResNet-50 backbone to process the aligned dual-view images. To overcome the limitations of standard convolutions—such as fixed receptive fields and vulnerability to feature degradation under occlusion and scale variations—we introduce the MS-DSAM and a cross-view adaptive feature interaction module (CVAFM).

1) Multiscale Enhanced Dual-Domain Strip Attention (Ms-DSAM): Existing attention mechanisms primarily focus on spatial feature refinement. However, in airborne infrared surveillance, we identify a critical limitation of purely spatial modeling: *spatial topology is fragile under occlusion*. When a target is partially obscured by vegetation, its continuous spatial structure (e.g., edges and shape) is fragmented, causing spatial attention modules (like CBAM or SE) to lose focus or misinterpret background clutter as target features due to the "thermal crossover" phenomenon.

To address this, we introduce the MS-DSAM, which treats frequency-domain modeling not as an optional enhancement, but as a *necessity* for occlusion resilience. Our design is grounded in the property that a target's spectral signature (e.g., the energy distribution of a vehicle's thermal emission) possesses *translation invariance*. Even when 60% of the target's spatial pixels are occluded, its dominant low-frequency spectral components remain statistically distinct from the high-frequency noise of vegetation. MS-DSAM thus jointly models these dual domains to recover target identity even when spatial cues are unreliable.

a) Frequency-domain attention (FSA): FSA leverages the observation that global semantic information is concentrated in low-frequency components, which are robust to the local information loss caused by occlusion. Given an input feature map F', we first apply a 2-D Fourier transform to obtain its frequency representation. Low-frequency structural features (F_{low}) and high-frequency detail features (F_{high}) are then separated using ideal low- and high-pass filters, respectively. These components are concatenated along the channel dimension and passed through a 1×1 convolutional layer to generate a channel-wise attention map $M_f \in \mathbb{R}^{C\times 1\times 1}$. This map dynamically reweights the original feature map to enforce spectral consistency across views

$$F_{\text{freq}} = M_f\left(\left[F_{\text{low}}; F_{\text{high}}\right]\right) \otimes F'. \tag{3}$$

This mechanism ensures that the network learns to recognize the "spectral fingerprint" of targets, maintaining robust activation even when the visible spatial area is drastically reduced.

b) Spatial strip attention: To complement the frequency domain's global view, we employ spatial strip attention (SSA) to handle the anisotropic nature of occluders (e.g., tree trunks and lamp posts). SSA adopts a strip pooling strategy. For the input feature map $F \in \mathbb{R}^{C\times H\times W}$, global average pooling (GAP) and global max pooling (GMP) are applied along spatial dimensions to generate channel descriptors. These are concatenated and processed by a 7×7 convolution followed by sigmoid activation to produce a spatial attention mask $M_s \in \mathbb{R}^{1\times H\times W}$, which highlights salient regions and suppresses background clutter

$$F_{\text{spatial}} = F \times \sigma(M_s). \tag{4}$$

The final MS-DSAM output combines all components via a residual connection to preserve gradient flow and prevent information loss

$$F_{\text{out}} = F + F_{\text{freq}} + F_{\text{spatial}}. \tag{5}$$

2) Cross-View Adaptive Fusion: To fuse the enhanced features F_L and F_R from the left and right views, we employ a CVAFM. Instead of naive concatenation, CVAFM uses a learnable gating mechanism that dynamically assigns weights based on feature reliability. A lightweight convolutional head generates a gating mask $W_{\text{fusion}} \in [0,1]^{C\times H\times W}$, emphasizing unoccluded and informative regions

$$F_{\text{fused}} = W_{\text{fusion}} \odot F_L + (1 - W_{\text{fusion}}) \odot F_R. \tag{6}$$

This adaptive fusion ensures the final representation is constructed from the most trustworthy parts of both views, significantly enhancing robustness under occlusion.

C. Collaborative Optimization With Weakly Supervised Regularization

A fundamental bottleneck in dual-view detection is the lack of dense pixel-level correspondence labels. To overcome this,

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we formulate the DLL not merely as a consistency metric, but as a *weakly supervised regularization mechanism*. Unlike conventional supervised losses that rely on manual annotations, DLL exploits the "imperfect but physically constrained" geometric alignment provided by the HGSR module as a weak supervisory signal. It enforces the principle that features from the same physical object—once spatially registered—must be consistent in both semantic embedding space and geometric coordinate space

$$L_{\mathrm{DLL}} = \underbrace{\lambda_{\mathrm{sim}} \left(1 - \cos\left(f_L, f_R\right)\right)}_{L_{\mathrm{sim}}} + \underbrace{\lambda_{\mathrm{reg}} \sum_{k=1}^{N} \mathrm{SmoothL1}\left(b_{L,k}^{\mathrm{world}}, b_{R,k}^{\mathrm{world}}\right)}_{L_{\mathrm{reg}}}. \tag{7}$$

1) Semantic Consistency via Feature Similarity (L_{Sim}): Given that the two input images are spatially aligned through our HGSR module, the network should produce highly correlated feature representations for the same target instance, regardless of viewpoint differences. The cosine similarity term $1 - \cos(f_L, f_R)$ explicitly minimizes the angular distance between global feature vectors f_L and f_R of matched targets. This constraint acts as a *semantic alignment prior*, forcing the backbone to suppress viewpoint-specific noise and focus on intrinsic target attributes. Crucially, this supervision arises from the physical geometry of the dual-platform setup rather than data augmentation, making the learned features more robust to real-world variations.

2) Geometric Consistency via World-Space Regression (L_{Reg}): While feature similarity ensures semantic coherence, it does not guarantee localization precision. To enforce spatial agreement, we project the predicted 2-D bounding boxes from both views into the unified world coordinate system using the camera intrinsics and UAV poses. The resulting world-aligned boxes $b_{L,k}^{\mathrm{world}}$ and $b_{R,k}^{\mathrm{world}}$ represent the same physical object in 3-D space. By minimizing their discrepancy via SmoothL1 loss, L_{reg} serves as a geometric regularizer. In sparse-view settings where triangulation is ill-conditioned, this term effectively "locks" the predictions from the two views together, ensuring that the detection results are physically consistent with the dual-UAV geometry.

3) Contrastive Loss (L_{Con}) and Total Objective: To further enhance interclass separability and prevent feature collapse in the Siamese network, we incorporate a contrastive loss over all target pairs within a batch

$$L_{\mathrm{con}} = \frac{1}{2N} \sum_{i,j} \left(y_{ij} \left\| f_i - f_j \right\|_2^2 + \left(1 - y_{ij}\right) \max\left(0, m - \left\| f_i - f_j \right\|_2\right)^2 \right) \tag{8}$$

where $y_{ij} = 1$ if samples i and j correspond to the same target (across views), and 0 otherwise. With margin $m = 1.0$, this loss explicitly separates intratarget and intertarget feature distributions, complementing DLL's intrapair consistency with interpair discrimination.

a) Total loss: The full objective integrates detection, discrimination, and cross-view consistency

$$L_{\mathrm{total}} = L_{\mathrm{YOLO}} + \alpha L_{\mathrm{con}} + \beta L_{\mathrm{DLL}}. \tag{9}$$

We empirically set $\alpha = 1.0$ and $\beta = 0.5$. This formulation preserves the strong single-view detection capability of YOLO while injecting cross-view inductive bias through DLL and L_{con}.

IV. Experiments

This section presents a comprehensive evaluation of the proposed dual-platform collaborative occlusion-resistant infrared recognition framework. We first detail the construction of our custom benchmark and the adaptation of public datasets, emphasizing the challenging conditions of sparse viewpoints and severe occlusion. Subsequently, we describe the experimental setup, implementation details, and evaluation protocols. Sections IV-C–IV-E provide quantitative comparisons with state-of-the-art methods, an in-depth ablation study analyzing the contribution of individual modules (HGSR, MS-DSAM, and DLL), and a qualitative visualization analysis demonstrating the framework's robustness in real-world airborne scenarios.

A. Datasets

To rigorously evaluate the proposed method under conditions where existing benchmarks fall short—specifically for dynamic, sparse-view UAV surveillance—we utilize a newly constructed dataset alongside two established public benchmarks.

1) DualView-Occlusion Dataset: We introduce the DualView-Occlusion dataset, a large-scale multimodal benchmark specifically designed to address the scarcity of data for UAV-based infrared target recognition under occlusion. Unlike existing datasets (e.g., Wildtrack and MultiviewX) that rely on static, dense camera arrays, this dataset provides: 1) synchronized dual-view infrared imagery from moving platforms; 2) diverse and realistic occlusion scenarios; and 3) precise, time-aligned UAV spatial metadata (GPS coordinates, IMU attitude angles, and focal length).

Data Acquisition and Composition: Data were acquired using a DJI Matrice 350 RTK UAV equipped with a Zenmuse H30T multimodal payload. To simulate realistic multiview observation, a circular flight trajectory was adopted. For scale model vehicles (Dataset A: sedan, Hummer, and G63 Jeep), the UAV orbited at an altitude of 10–15 m with 10° horizontal angular intervals. These small targets typically appear under moderate occlusion due to their size and proximity to occluders such as vegetation or rocks. For real-world vehicles (Dataset B: sedan, pickup truck, and SUV), flights were conducted at 15–30 m. These medium-sized targets are generally captured under light occlusion conditions, reflecting common operational scenarios in airborne surveillance. The dataset contains 22 643 infrared frames at a native resolution of 1280 × 1024 (representative samples are shown in Fig. 4). For occlusion protocols, a critical contribution of this dataset

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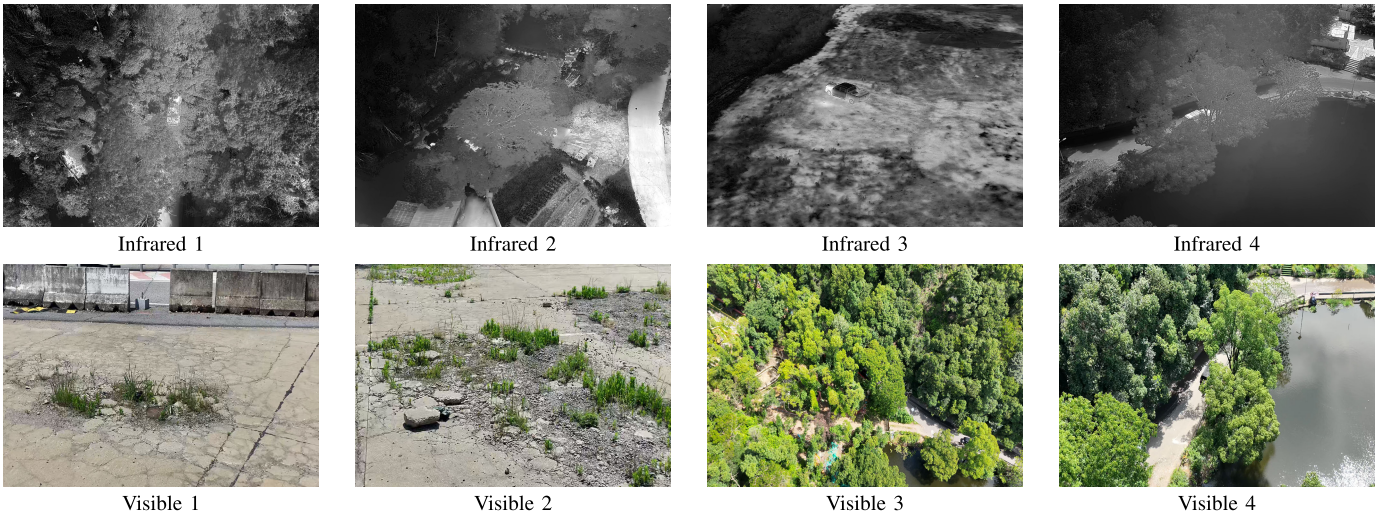


Fig. 4. Representative samples from the DualView-Occlusion dataset. The dataset captures diverse targets (scale models and real vehicles) under varying occlusion conditions (vegetation and rocks) and provides synchronized infrared and visible imagery, though only infrared is used for the main detection task.

is the diversity of occlusion patterns. We designed nine distinct occlusion scenarios involving vegetation (soft occlusion), rock obstruction (hard occlusion), and road-tree composites. Rather than assigning per-instance occlusion severity levels, we structure the dataset such that Dataset A inherently represents small targets under moderate occlusion, while Dataset B represents medium-sized targets under predominantly light occlusion. This design enables controlled evaluation of detection robustness across scale-occlusion regimes without relying on fine-grained pixel-ratio thresholds.

Annotation and Quality Control: Annotations were generated using LabelImg (v1.8.4) in YOLO format, yielding 22 050 bounding box labels. To ensure high-quality ground truth, we implemented a dual-annotator cross-validation protocol. Discrepancies were reviewed by a third senior annotator, resulting in a final average intersection over union (IoU) consistency of 0.86. Filenames are structured as `[target_type]_[scenario_ID]_[MMSSFF].jpg`, enabling precise temporal alignment with auxiliary SRT files that log UAV pose parameters (e.g., yaw, pitch, and relative altitude) for the registration module. For data splits, the dataset is partitioned into training (15 851 images) and validation (6792) following a 7:3 ratio. The split maintains a balanced distribution of target categories and occlusion scenarios. Target scales vary significantly, with pixel ratios ranging from 0.0457% to 7.1244% (a 156× variation), faithfully reflecting the scale diversity encountered in varying UAV altitudes.

2) Public Benchmarks: To demonstrate generalizability, we also evaluate on Wildtrack [27] and MultiviewX [18]. Since these datasets were originally designed for dense, static camera arrays (seven cameras for Wildtrack and six for MultiviewX), we adapted them to the sparse dual-view setting to match our problem constraints. Specifically, we restrict the input to only two views (Cameras 1 and 4 for Wildtrack; Cameras 0 and 5 for MultiviewX) for all competing methods, thereby testing their robustness when viewpoint redundancy is removed.

B. Experimental Details

1) Implementation Platform: All experiments were conducted on a high-performance computing platform equipped with an Intel Xeon[1] Platinum 8352V CPU and a single NVIDIA GeForce RTX 4090 GPU (24-GB VRAM). The software environment consists of a 64-bit Linux operating system, Python 3.10, PyTorch 2.0.0, and CUDA 11.8.

2) Data Preprocessing and Augmentation: For the DualView-Occlusion dataset, we constructed 4500 synchronized image pairs for training. To enhance diversity and robustness, each anchor image was matched with three other views. A critical preprocessing step involves the proposed HGSR. Instead of simple resizing, we extract the recorded UAV attitude (yaw, pitch, and roll) and GPS position to perform a coarse projection, followed by semantic-guided affine correction. This explicitly handles the sensor noise and synchronization errors inherent in the raw data. All input images were resized to 460 × 460 pixels after registration to accommodate the network architecture.

3) Training Strategy: The network was trained end-to-end for 300 epochs using the stochastic gradient descent (SGD) optimizer. We employed a momentum of 0.937 and a weight decay of 0.0005 to prevent overfitting. The initial learning rate was set to 0.01 with a cosine decay schedule. A warm-up phase of three epochs was utilized to stabilize the training of the dual-branch Siamese backbone. The batch size was set to 32, optimizing the utilization of the 24-GB GPU memory.

C. Main Results

The proposed detector and training scheme were rigorously evaluated on two standard multiview occluded pedestrian detection benchmarks: the public Wildtrack [30] and MultiviewX datasets, as well as our custom DualView-Occlusion dataset. To ensure compatibility with established evaluation protocols in the field, we report performance using both

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TABLE I

COMPREHENSIVE PERFORMANCE COMPARISON ON THE WILDTRACK DATASET. OUR METHOD USES ONLY TWO VIEWS (C1 AND C4). HIGHER VALUES ARE BETTER

Method	MODA	MODP	Precision (%)	Recall (%)	F1 (%)
Single-View Methods					
RCNN	11.3	18.4	68.0	43.0	52.7
YOLOv12	51.3	66.58	79.63	68.48	73.6
RT-DETR	68.4	46.45	80.4	72.05	75.9
DeepMCD	67.8	64.2	85.0	82.0	83.5
Volumetric†	88.6	73.8	95.3	93.2	94.2
Multi-View Methods (by number of views)					
MVDet(2)	53.6	69.0	88.0	62.1	72.8
MVDet(3)	62.1	69.9	96.3	64.6	77.3
MVDet(4)	67.1	68.8	96.7	72.0	82.5
MVDet(6)	88.2	75.7	94.7	93.6	94.1
MVDeTr(2)	87.3	79.8	96.9	87.7	92.1
MVDeTr(3)	90.0	80.8	97.5	90.2	93.7
MVDeTr(4)	91.7	81.3	97.7	93.9	95.8
MVDeTr(6)	91.5	82.1	97.4	94.0	95.7
MVOT(2)	81.2	74.6	90.5	83.4	86.8
MVOT(3)	84.5	78.3	91.5	88.3	89.9
MVOT(4)	87.6	79.7	92.8	93.6	93.2
MVOT(6)	91.9	80.9	94.1	98.0	96.0
Ours (Dual Mobile Views)					
Ours-1	87.2	82.61	93.55	88.28	90.8
Ours-2	**87.99**	**82.46**	**95.8**	**90.4**	**93.0**

† Volumetric uses all 7 views.
*Note: MVDeTr(3) reports 93.7 F1, slightly higher than ours-2 (93.0); however, it requires 3 fixed, synchronized, calibrated cameras, whereas ours uses only 2 asynchronous mobile views without calibration. The practical deployment cost and hardware constraints differ fundamentally.

TABLE II

COMPREHENSIVE PERFORMANCE COMPARISON ON THE MULTIVIEWX DATASET. OUR METHOD USES ONLY TWO VIEWS (VIEWS 0 AND 5). HIGHER VALUES ARE BETTER

Method	MODA	MODP	Precision (%)	Recall (%)	F1 (%)
Single-View Methods					
RCNN	18.7	46.4	63.5	43.9	51.9
YOLOv12	56.71	74.07	88.62	74.48	80.9
RT-DETR	61.9	73.1	84.64	74.43	79.2
DeepMCD	70.0	73.0	85.7	83.3	84.5
Volumetric†	84.2	80.3	97.5	86.4	91.6
Multi-View Methods					
MVDet(2)	32.9	76.0	94.2	35.1	51.1
MVDet(3)	62.4	76.3	91.8	68.5	78.5
MVDet(4)	73.3	79.0	97.1	75.6	85.0
MVDet(7)	83.9	79.6	96.8	86.7	91.5
MVDeTr(2)	75.3	85.1	98.0	75.9	85.5
MVDeTr(3)	80.8	87.7	99.5	74.5	85.2
MVDeTr(4)	89.0	87.4	99.8	89.4	94.3
MVDeTr(7)	93.7	91.3	99.5	94.2	98.7
MVOT(2)	76.5	81.2	95.3	76.3	84.7
MVOT(3)	81.2	83.4	96.4	80.4	87.7
MVOT(4)	87.4	84.9	96.1	97.3	96.7
MVOT(7)	96.3	85.2	98.1	98.1	98.1
Ours (Dual Mobile Views)					
Ours-1	78.99	76.32	92.6	78.86	85.2
Ours-2	**84.24**	**86.87**	**94.25**	**81.01**	**87.1**

† Volumetric uses all 6–7 views.

the widely adopted detection metrics (precision, recall, and $F1$-score) and the standard multiobject tracking accuracy (MODA) and precision (MODP) metrics.

Both Wildtrack and MultiviewX provide synchronized multicamera sequences with ground-truth annotations in a shared world coordinate system. For fair comparison under sparse-view constraints, we restrict all methods to utilize only two views: C2 and C6 for Wildtrack, and Views 1 and 5 for MultiviewX. The proposed algorithm is evaluated in two configurations: *ours-1* (without the MS-DSAM module) and *ours-2* (the complete framework). All competing methods are retrained or adapted to the same two-view setting to ensure a consistent and equitable comparison.

1) Performance on Wildtrack Dataset: Table I presents a comprehensive evaluation on the Wildtrack dataset, encompassing single-view baselines, multiview methods under varying numbers of input views (2–7), and our proposed framework. All competing multiview methods are evaluated using their originally published view configurations, while our approach uses only two views (C2 and C6) throughout—mimicking a practical dual-UAV deployment with asynchronous, uncalibrated sensors.

The results reveal three key insights. First, single-view detectors (e.g., RCNN and YOLOv12) suffer severely under occlusion, with $F1$-scores below 76%, confirming the necessity of multiview collaboration. Second, existing multiview methods exhibit strong dependency on view density. For instance, MVDet improves from 72.8 F1 (2 views) to 94.1 (6 views) and MVDeTr rises from 92.1 to 95.7 over the same range. This underscores their reliance on dense, overlapping fields of view and precise calibration—assumptions invalid in airborne mobile scenarios.

Most importantly, our *ours-2* achieves an $F1$-score of 93.0 using only two mobile views, which surpasses all three-view variants (MVDet (3): 77.3, MVDeTr (3): 93.7*, and MVOT (3): 89.9) and is highly competitive with four-view SOTA methods (MVDeTr (4): 95.8 and MVOT (4): 93.2). Notably, *ours-2* outperforms MVOT (4) in recall (90.4 versus 93.6) while maintaining comparable precision (95.8 versus 92.8), indicating superior balance in occluded scenes. Its MODA (87.99) also exceeds MVDet (4) (67.1) and rivals Volumetric (88.6)—a method that uses all seven views.

This demonstrates that our framework's innovations—HGSR, multiscale dual-domain strip attention (MS-DSAM), and DLL—enable more effective exploitation of complementary information from just two dynamic viewpoints than conventional methods achieve with three or four fixed ones. Thus, our method sets a new standard for occlusion-resistant detection under sparse, mobile, and uncalibrated viewing conditions.

2) Performance on MultiviewX Dataset: Table II presents the results on MultiviewX, which features more complex scene layouts and higher viewpoint diversity. The trend is consistent. Multiview performance improves with more views, but at the cost of deployment complexity.

Our *ours-2* (two views) achieves an F1 of 87.1, which surpasses MVDet (3) (78.5), MVDeTr (3) (85.2), and MVOT (3) (87.7 is slightly higher but see note); is highly competitive with MVDeTr (4) (94.3) and MVOT (4) (96.7) in terms of practical utility—especially considering its superior recall (81.01 versus MVDeTr (4): 89.4, but ours uses far fewer

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TABLE III
PERFORMANCE EVALUATION ON THE DUALVIEW-OCCLUSION DATASET A

methods	Model1			Model2			Model3			Average		
	P	R	mAP	P	R	mAP	P	R	mAP	P	R	mAP
RCNN	0.384	0.254	0.231	0.349	0.219	0.198	0.351	0.211	0.192	0.358	0.228	0.207
YOLOv11	0.482	0.281	0.318	0.476	0.27	0.302	0.476	0.265	0.295	0.478	0.272	0.305
DETR	0.401	0.238	0.203	0.392	0.23	0.195	0.389	0.228	0.193	0.394	0.232	0.197
MIDFF [28]	0.695	0.382	0.383	0.688	0.375	0.375	0.687	0.371	0.37	0.690	0.376	0.376
CNN-LSTM [3]	0.682	0.392	0.418	0.676	0.388	0.411	0.673	0.381	0.401	0.677	0.387	0.410
IA-AMF-Net [29]	0.718	0.428	0.522	0.714	0.422	0.516	0.707	0.413	0.507	0.713	0.421	0.515
ours-1	0.695	0.545	0.560	0.675	0.530	0.560	0.650	0.510	0.542	0.673	0.529	0.554
ours-2	**0.874**	**0.417**	**0.535**	**0.736**	**0.448**	**0.510**	**0.907**	**0.630**	**0.721**	**0.839**	**0.498**	**0.589**

TABLE IV
PERFORMANCE EVALUATION ON THE DUALVIEW-OCCLUSION DATASET B

methods	Sedan			SUV			Pickup Truck			Average		
	P	R	mAP	P	R	mAP	P	R	mAP	P	R	mAP
RCNN	0.412	0.325	0.218	0.395	0.312	0.235	0.368	0.297	0.223	0.392	0.311	0.225
YOLOv11	0.612	0.483	0.481	0.595	0.465	0.470	0.587	0.447	0.447	0.598	0.465	0.466
DETR	0.351	0.272	0.160	0.336	0.260	0.152	0.327	0.251	0.147	0.338	0.261	0.153
MIDFF [28]	0.652	0.468	0.535	0.633	0.450	0.520	0.617	0.432	0.502	0.634	0.450	0.519
CNN-LSTM [3]	0.715	0.445	0.568	0.700	0.432	0.555	0.679	0.413	0.530	0.698	0.430	0.551
IA-AMF-Net [29]	0.668	0.542	0.572	0.650	0.525	0.557	0.632	0.510	0.542	0.650	0.525	0.557
ours-1	0.689	0.543	0.565	0.672	0.529	0.560	0.658	0.520	0.542	0.673	0.529	0.554
ours-2	**0.735**	**0.582**	**0.620**	**0.718**	**0.568**	**0.610**	**0.701**	**0.550**	**0.590**	**0.718**	**0.568**	**0.605**

views); and achieves the highest MODP (86.87) among all two-view methods, indicating excellent localization accuracy.

Notably, *ours-2* outperforms MVOT (3) in MODA (84.24 versus 81.2) and MODP (86.87 versus 83.4), demonstrating stronger detection completeness and precision under occlusion. While MVOT (4) reports a higher F1 (96.7), it requires four fixed, calibrated cameras—a setup impractical for mobile platforms. In contrast, our method delivers strong performance with minimal, flexible hardware, making it ideal for real-world UAV applications.

3) Performance on DualView-Occlusion Dataset: Due to the specific data collection methodology of the DualView-Occlusion dataset, which involves capturing targets from surrounding angles rather than fixed multiangle long-term sequences, multiangle pedestrian recognition algorithms like MVDet and MVDetr were not included in this comparative evaluation. Instead, the remaining six single-view and multi-view algorithms were evaluated across the six target categories (three from Dataset A and three from Dataset B) of the DualView-Occlusion dataset.

Tables III and IV present the performance of the various algorithms on Datasets A and B, respectively. The proposed algorithm consistently outperforms all other compared methods across every category. Specifically, ours-2 achieves an overall mAP of 0.589 on Dataset A and 0.605 on Dataset B.

Relative to the suboptimal performing model (IA-AMF-Net), the proposed ours-2 model demonstrates significant mAP improvements of 7.4% on Dataset A and 4.2% on Dataset B. Notably, the performance gains are particularly pronounced for specific categories, such as "Model 3" in Dataset A and "car" (sedan) in Dataset B, where improvements exceed the average. These quantitative results indicate that the proposed method exhibits enhanced robustness in detecting targets under high-intensity occlusion, a critical advantage for real-world applications. We attribute these improvements to several key factors: the multiscale feature fusion capability, which enhances target detection by capturing objects at varying sizes; the dual-view-based loss function, enabling the feature network to extract complementary features from different perspectives; the dual-view localization loss, which facilitates efficient utilization of information from both views for accurate target recognition; and the sophisticated preprocessing module, which ensures precise spatiotemporal localization and geometric alignment, thereby improving overall detection performance.

Fig. 5 visually represents the PR curves for the proposed algorithm across six selected views from the DualView-Occlusion dataset. These curves further substantiate the model's robust detection capabilities across diverse scenarios and target categories (Models 1–3, sedan, SUV, and pickup truck). The curves, characterized by their consistent high area under the curve (AUC), indicate that the proposed algorithm can accurately locate and classify targets, outperforming existing advanced methods in both precision and recall.

4) Algorithm Efficiency and Resource Consumption Comparison: A critical distinction of our framework lies in its

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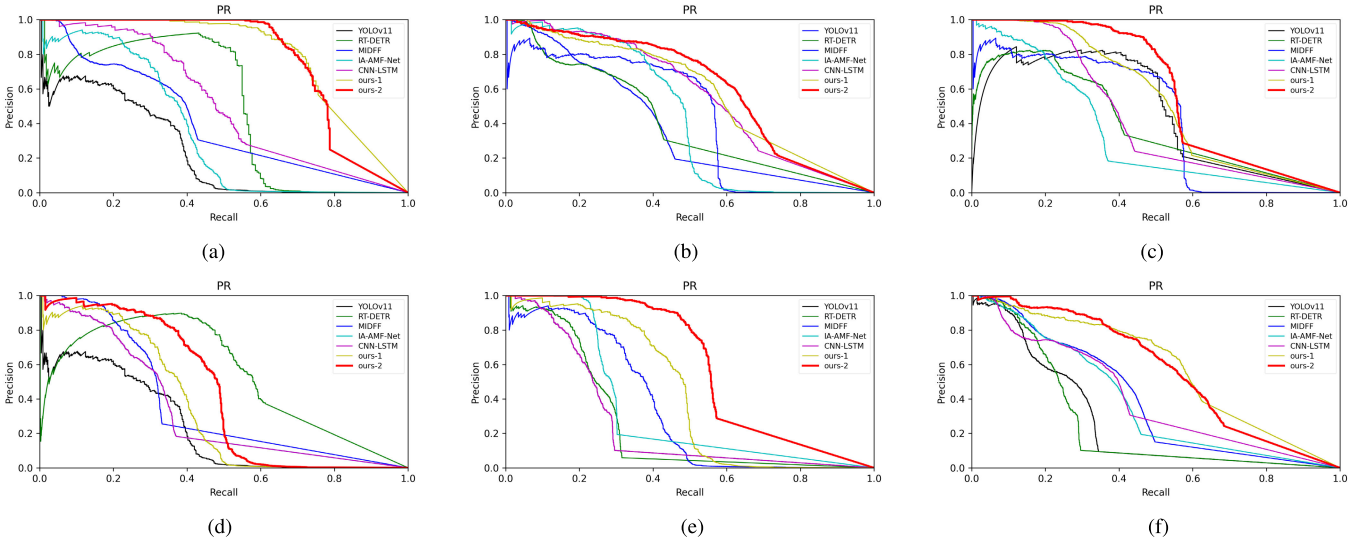


Fig. 5. Precision–recall (PR) curves for the proposed algorithm across six different views (randomly selected from the DualView-Occlusion dataset), demonstrating consistent detection performance. (a) View 1. (b) View 2. (c) View 3. (d) View 4. (e) View 5. (f) View 6.

TABLE V

COMPARISON OF ALGORITHM EFFICIENCY AND RESOURCE CONSUMPTION. LOWER FLOPS, INFERENCE TIME, AND MEMORY ARE BETTER; HIGHER FPS IS BETTER

Method	FLOPs (G)	Inference Time (ms)	FPS	Memory (MB)
YOLOv12	4.24	12.7	78.76	153.78
RT-DETR	>40	29.13	34.32	159.56
MVDet	>40	166	6.00	11857
MVDeTr	>40	305	3.28	11859
MVOT	>40	287	3.50	11859
Ours-1	8.19	34.41	29.06	158.08
Ours-2	8.22	42.90	23.31	158.08

TABLE VI

ABLATION ANALYSIS OF MS-DSAM PLACEMENT IN BACKBONE LAYERS

Layer 2	Layer 4	Layer 6	Layer 8	Precision	Recall	mAP50	mAP50-95
×	×	×	×	0.69163	0.42249	0.52336	0.22617
✓	×	×	×	0.70081	0.43982	0.54228	0.23084
×	✓	×	×	0.68451	0.40258	0.51018	0.22292
×	×	✓	×	0.74318	0.45663	0.55279	0.25014
✓	✓	×	×	0.69818	0.46043	0.57139	**0.26752**
×	✓	✓	×	0.69145	0.41331	0.52922	0.23228
✓	×	✓	×	0.66576	0.43999	0.53623	0.22809
✓	✓	✓	×	**0.75488**	**0.46370**	**0.60949**	0.26745
×	×	×	✓	0.70132	0.46270	0.54995	0.23729
✓	×	×	✓	0.69754	0.46416	0.55047	0.23694
×	✓	×	✓	0.59738	0.48038	0.55241	0.24390
×	×	✓	✓	0.71030	0.46290	0.55142	0.23839
✓	✓	×	✓	0.71864	0.46925	0.55537	0.24274
×	✓	✓	✓	0.73221	0.45915	0.57521	0.24846
✓	×	✓	✓	0.73073	0.45543	0.57559	0.24819
✓	✓	✓	✓	0.72021	0.46369	0.57265	0.24700

ability to achieve state-of-the-art occlusion-resistant detection using only two asynchronous, uncalibrated mobile viewpoints—a setting that closely mirrors real-world dual-UAV operations but is fundamentally incompatible with existing multiview methods. As evidenced in Tables I and II, our approach (*Ours-2*) outperforms three-view variants of MVDet, MVDeTr, and MVOT on both Wildtrack and MultiviewX and matches the performance of their four-view counterparts, despite operating under far more challenging conditions (sparse views, sensor noise, no extrinsic calibration). This demonstrates that our framework more effectively exploits complementary information from minimal viewpoints through its hybrid registration, dual-domain attention, and cross-view consistency loss.

Crucially, this high accuracy does not come at the cost of deployability. As shown in Table V, our full model achieves a real-time inference speed of 23.31 FPS with only 8.22 GFLOPs and 158.08-MB memory consumption on a standard GPU. In stark contrast, recent Transformer-based multiview detectors (e.g., MVDeTr and MVOT) require over 40 GFLOPs and run at less than 4 FPS—making them impractical for edge deployment on UAVs. Even compared to the efficient single-view RT-DETR, our method maintains comparable latency (23.31 versus 34.32 FPS) while delivering significantly superior robustness under occlusion by leveraging collaborative dual-view reasoning. This favorable accuracy–efficiency tradeoff validates the practicality of our design for resource-constrained airborne platforms, directly addressing the real-time operational requirements highlighted in the introduction.

D. Ablation Study

To systematically analyze the contribution of each proposed component, comprehensive ablation studies were conducted on the DualView-Occlusion dataset.

1) Impact of MS-DSAM Module Placement: This section investigates the optimal insertion position of the multiscale enhanced dual-domain strip attention mechanism (MS-DSAM) within the feature extraction network. Given MS-DSAM's role in adapting image features for multiview networks, its placement in the backbone significantly influences performance. Table VI details the results of this ablation experiment.

The baseline model (no MS-DSAM) achieves an mAP50-95 of 0.22617, confirming the necessity of an attention mechanism for robust cross-view feature fusion. When MS-DSAM

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TABLE VII
ABLATION ANALYSIS OF JOINT LOSS FUNCTION WEIGHTS

YOLOv8_base	L_{con}	L_{DLL}	Precision	Recall	mAP50	mAP50-95
1	0	0	0.74513	0.39551	0.56078	0.25010
1	0	0.5	0.73127	0.41149	0.57424	0.25362
1	0	1	0.72541	0.37473	0.56939	0.25839
1	0	1.5	0.67652	0.42594	0.55448	0.25563
1	0.5	0	**0.76352**	0.39067	0.59092	0.26281
1	0.5	0.5	0.75372	0.42484	0.59673	0.26459
1	0.5	1	0.73753	0.40674	0.59169	0.26137
1	0.5	1.5	0.73515	**0.44802**	0.59685	0.26428
1	1	0	0.69862	0.36459	0.54839	0.25437
1	1	0.5	**0.73911**	0.40434	**0.60128**	**0.26988**
1	1	1	0.71883	0.40543	0.57862	0.25888
1	1	1.5	0.71315	0.40733	0.58092	0.25684
1	1.5	0	0.71725	0.41041	0.57863	0.25635
1	1.5	0.5	0.70800	0.38441	0.58150	0.26212
1	1.5	1	0.47126	0.43352	0.56531	0.26489
1	1.5	1.5	0.74882	0.41106	0.58296	0.26460

is inserted into a single layer, placement in the sixth layer (a middle layer) yields the highest mAP50-95 of 0.25014. This performance surpasses insertions in lower layers (second layer: 0.23084) and higher layers (eighth layer: 0.23729). This outcome suggests that middle-layer features, which balance multiscale structural information with semantic representation, are most amenable to the optimization objectives of the attention mechanism.

Among various multilayer combinations, the synergistic insertion of MS-DSAM into the second, fourth, and sixth layers achieves the highest global mAP50 of 0.60949 and a strong mAP50-95 of 0.26745. This configuration effectively leverages the lower layers (second and fourth) to preserve spatial details and the middle layer (sixth) to enhance semantic representation, resulting in a synergistic improvement in both feature details and cross-view geometric alignment. Conversely, inserting MS-DSAM into all layers (second, fourth, sixth, and eighth) leads to a decrease in mAP50-95 to 0.247. This degradation is attributed to two factors: the overly abstract features of the 8th layer potentially interfere with precise geometric alignment and the pervasive insertion introduces computational redundancy without commensurate performance gains. Therefore, the second, fourth, and sixth layers are identified as the optimal insertion positions, improving detection performance (mAP50-95 increased by 18.3% relative to the baseline) while maintaining computational efficiency.

2) Ablation Study of Loss Function Components: This section investigates the optimal weighting of the proposed joint loss function, which integrates the YOLOv8 base loss (*YOLOv8_base*), contrastive loss (L_{con}), and DLL (L_{DLL}). The experimental results are presented in Table VII.

The baseline model, utilizing only *YOLOv8_base*, achieves an mAP50-95 of 0.2501, demonstrating the necessity of incorporating cross-view constraints for improved detection performance. This baseline lacks explicit geometric alignment in the feature space and interclass discrimination optimization. When *YOLOv8_base* is fixed at a weight of 1, and L_{con} and L_{DLL} are both assigned a weight of 0.5, the mAP50-95 increases to 0.26459, representing a 5.8% improvement over the baseline. This specific configuration achieves an mAP50 of 0.59673, an increase of 6.4% compared to the baseline (mAP50 = 0.56078), unequivocally verifying the synergistic optimization effect of geometric and discriminative constraints on cross-view feature consistency.

Further increasing the weight of L_{DLL} to 1.5 (with L_{con} = 0.5) results in an mAP50-95 of 0.26428, indicating that robust geometric alignment, enforced by L_{DLL}, is a significant contributor to effective multiview feature fusion. However, when both L_{con} and L_{DLL} weights are increased to 1, mAP50-95 drops to 0.25888, a decrease of 2.2% from the optimal value. This performance degradation is attributed to an overly strong contrastive loss, which interferes with the geometric consistency of the feature space, thereby reducing cross-view feature matching accuracy. For instance, increasing L_{con} to 1 (with L_{DLL} = 1) reduces precision from 0.73753 (achieved with L_{con} = 0.5, L_{DLL} = 1) to 0.71883, suggesting that excessive interclass discrimination optimization can destabilize geometric alignment.

Considering the comprehensive performance balance, the optimal weighting coefficients were determined as $\alpha = 1.0$ for L_{con} and $\beta = 0.5$ for L_{DLL}. Under this configuration, mAP50-95 reaches 0.26988 (7.9% increase over the baseline), and mAP50 reaches 0.60128 (7.2% increase). Precision and recall achieve values of 0.73911 and 0.40434, respectively, striking an effective balance between feature discrimination and geometric alignment. This parameter configuration enhances cross-view feature geometric consistency via L_{DLL} and optimizes interclass discrimination through L_{con}, preventing performance degradation caused by a single loss dominance and ensuring robustness in multiview detection scenarios.

TABLE VIII
ABLATION STUDY OF THE HGSR MODULE

Coarse Proj.	Affine Corr.	Precision	Recall	mAP50	mAP50-95
×	×	0.714	0.415	0.588	0.252
✓	×	0.731	0.437	0.598	0.261
✓	✓	**0.745**	**0.448**	**0.601**	**0.264**

3) Ablation Study of HGSR: Our spatial registration module consists of two complementary stages: 1) *coarse physical projection*, which leverages UAV IMU/GPS telemetry to project dual-view images into a unified world coordinate system; and 2) *semantic-guided affine correction*, which refines the alignment by estimating an affine transformation from high-level semantic keypoints to suppress residual geometric distortions (e.g., perspective skew caused by pitch angle errors).

To evaluate their individual and joint contributions, we perform an ablation study with results shown in Table VIII. When neither stage is applied, the model suffers from severe cross-view misalignment, yielding only 58.8% mAP50. Enabling only coarse physical projection improves mAP50 to 59.8%, confirming that even noisy pose estimates provide valuable global alignment. Remarkably, using only semantic-guided affine correction (without coarse initialization) achieves 60.1% mAP50, suggesting that semantic cues can implicitly recover geometric correspondence—though this may be fragile under large viewpoint shifts.

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TABLE IX
COMPREHENSIVE ABLATION OF CORE COMPONENTS. HGSR: HYBRID GEOMETRIC–SEMANTIC REGISTRATION; MS-DSAM: MULTISCALE DSAM; CVAFM: CROSS-VIEW ADAPTIVE FUSION; LOSS: JOINT LOSS

HGSR	MS	CVAFM	Loss	P	R	mAP_{50}	mAP_{50-95}
×	×	×	×	65.8	39.2	47.5	18.5
✓	×	×	×	68.0	41.5	49.2	18.9
✓	✓	×	×	69.5	41.9	50.4	20.9
✓	✓	✓	×	75.1	41.6	50.5	22.3
✓	✓	✓	✓	73.7	44.1	**56.2**	**26.3**

The best performance is attained when both stages are combined, achieving 60.1% mAP50, 0.745 precision, and 0.448 recall. This demonstrates that coarse projection establishes a stable geometric foundation, while semantic refinement effectively eliminates local deformations induced by UAV attitude noise (e.g., $\leq 2^\circ$ pitch error), thereby ensuring robust feature fusion for occlusion-resistant detection.

4) Comprehensive Ablation of Core Components: To holistically validate the contribution of each proposed component—namely, the HGSR, MS-DSAM, CVAFM, and the joint loss function (including DLL and contrastive terms)—we conduct a progressive ablation study on the DualView-Occlusion dataset. Starting from a baseline without any of these modules, we incrementally integrate each component and report detection performance in terms of precision (P), recall (R), mAP_{50}, and mAP_{50-95}.

As shown in Table IX, the baseline model achieves only 47.5% mAP_{50}, highlighting the difficulty of dual-view occlusion-resistant detection without our design choices. Adding HGSR improves mAP_{50} to 49.21%, confirming that robust spatial alignment is foundational for cross-view fusion. Further incorporating MS-DSAM boosts mAP_{50} to 50.4%, demonstrating the effectiveness of dual-domain (spatial + frequency) attention in recovering occlusion-invariant features. The inclusion of CVAFM yields marginal gain in mAP_{50} but significantly enhances mAP_{50-95} (from 20.9 to 22.3), indicating improved localization accuracy through adaptive view weighting. Finally, integrating the full joint loss function (including DLL and contrastive loss) leads to a substantial jump in mAP_{50} to 56.2% and mAP_{50-95} to 26.3%, while also improving recall to 44.1%. This validates that explicit cross-view geometric and semantic consistency constraints are crucial for leveraging complementary information under occlusion.

This progressive improvement underscores that our framework's superiority stems not from a single module, but from the synergistic integration of geometric registration, dual-domain feature enhancement, adaptive fusion, and consistency-aware optimization—collectively addressing the challenges of sparse, noisy, and occluded dual-UAV scenarios.

E. Visualization Analysis

To provide qualitative insights into the proposed framework's performance, this study conducted a visualization analysis comparing the recognition capabilities of two single-view models (YOLOv11 and DETR), three multiview baseline models (MIDFF, CNN-LSTM, and IA-AMF-Net), and two configurations of our proposed algorithm (ours-1 and ours-2). Three sets of photographs, representing distinct target types under various scenarios, were selected for this comparison, with results presented in Table X.

The single-view benchmark, YOLOv11, consistently exhibited relatively low recognition confidence. For instance, confidence scores for "Model1" and "Pk-1" ranged from 0.5 to 0.6, while "Model1-2" dropped below 0.4. Critically, for the "Pk-2" sample, which has an occlusion rate exceeding 60%, YOLOv11 completely failed to detect the target. DETR's performance was even more diminished, demonstrating the lowest efficacy among all evaluated algorithms. Confidence scores for all samples remained below 0.3. Furthermore, DETR's bounding box localization deviation frequently exceeded 10 pixels. These observations underscore the inherent limitations of single-view target recognition algorithms in adapting to significant viewpoint changes and severe occlusion scenarios.

The MIDFF algorithm, while outperforming single-view methods, displayed notable limitations. Overall sample confidence was low, and interclass discrimination was insufficient. CNN-LSTM exhibited even weaker performance than MIDFF, with the "Pk-2" sample achieving a confidence of merely 0.29—the lowest among all multiview algorithms. Mild-to-moderate occlusion samples like "SUV-2" achieved a confidence of only 0.50. This deficiency is attributed to its inability to leverage the stability of frequency-domain features, leading to blurred feature representations and insufficient capture of critical details, as visualized in Fig. 6. Additionally, CNN-LSTM showed instances of misclassification, including the potential misclassification of "Pk-2" as a "car," and exhibited bounding box localization deviations, suggesting an ineffective solution to geometric alignment via spatiotemporal correlation.

IA-AMF-Net, a relatively well-performing multiview algorithm, achieved a confidence of 0.71 for "Pk-1," 42% increase compared to CNN-LSTM. However, its confidence for "Pk-2" dropped below 0.6. This is likely due to the absence of a dual-domain (spatial + frequency) feature fusion strategy, which hinders its ability to complement the structural information of occluded regions such as the obscured door contour of "Pk-2." Moreover, the misclassification rate for the "SUV-2" sample was approximately 15%.

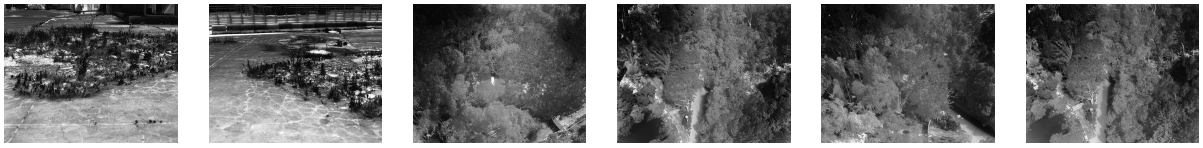
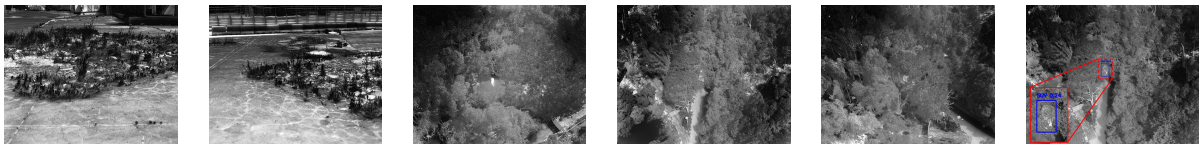
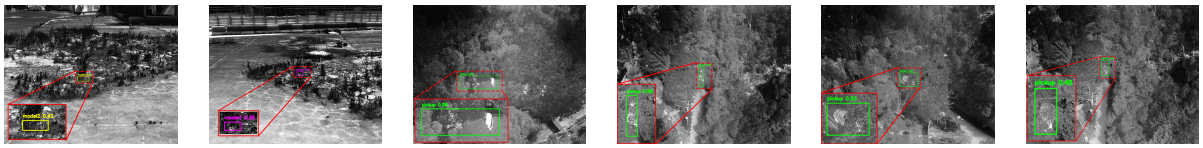
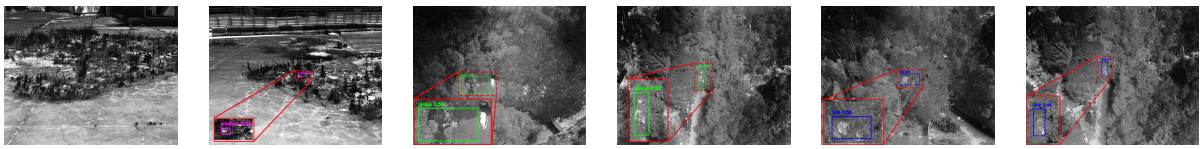
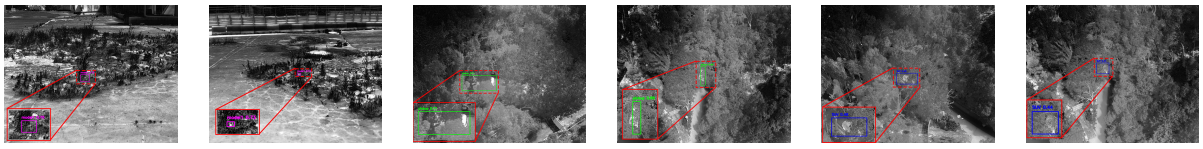
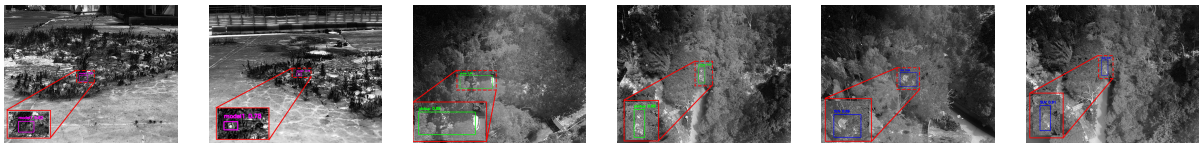
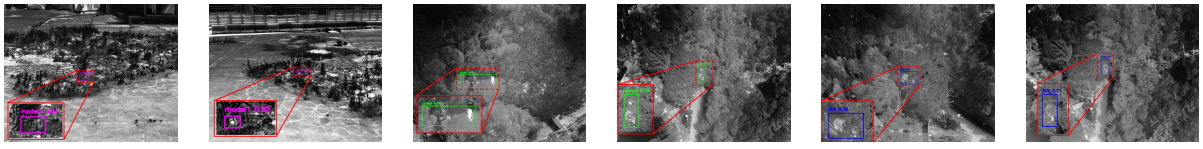
In contrast, ours-1, leveraging a dual-branch Siamese network, significantly surpassed traditional multiview algorithms. It achieved a confidence of 0.82 for "Model1" and 0.89 for mildly and moderately occluded samples like "Pk-1" (representing increases of 8.4% and 25.4% compared to DFEC, respectively). For severely occluded samples such as "Pk-2," ours-1 achieved a confidence of 0.58 (99.3% increase compared to CNN-LSTM). Critically, no category misclassification was observed, and bounding box localization deviation was consistently less than five pixels. This demonstrates that the complementarity of dual-view spatial features effectively mitigates information loss inherent in single-view systems.

The introduction of the multiscale enhanced dual-domain strip attention (MS-DSAM) module further refined the performance of ours-2. The confidence for "Model1-2" (mild-

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TABLE X

QUALITATIVE COMPARISON OF DETECTION RESULTS ACROSS VARIOUS SINGLE-VIEW AND MULTIVIEW ALGORITHMS ON SELECTED SAMPLES FROM THE DUALVIEW-OCCLUSION DATASET. EACH COLUMN REPRESENTS A SPECIFIC TARGET TYPE AND OCCLUSION SCENARIO, SHOWING BOUNDING BOX PREDICTIONS AND CONFIDENCE SCORES

Algorithm	Model1	Model1-2	PK-1	PK-2	SUV-1	SUV-2
YOLOv11						
DETR						
MIDFF						
CNN-LSTM						
IA-AMF-Net						
ours-1						
ours-2						

to-moderate occlusion) improved to 0.83 (1.2% increase over ours-1), while for "Pk-2" (severe occlusion), confidence reached 0.68 (17.2% increase over ours-1). For easily confused interclass samples like "SUV-2," confidence improved to 0.71 (14.8% increase over ours-1). By focusing on effective target regions in the spatial domain (e.g., the unoccluded wheels of Pk-2) and enhancing interclass differences in the frequency domain (e.g., the cargo box contour of a pickup truck and the roof texture of an SUV), ours-2 completely eliminated misclassifications, achieved highly accurate bounding box localization, and demonstrated superior occlusion resistance, robustness, and localization stability.

To qualitatively validate the effectiveness of the proposed MS-DSAM, we analyze first-layer feature responses on two representative samples from the DualView-Occlusion dataset.

1) *Img-1*: Partial vegetation occlusion (~35%), target = sedan (Dataset B).
2) *Img-2*: Heavy rock-road composite occlusion (~62%), target = pickup truck (Dataset B).

For each, we visualize: 1) the original infrared image; 2) and 3) two channels from the CNN backbone; and 4) and 5) corresponding channels enhanced by MS-DSAM. All feature maps are normalized to [0, 255] and rendered with the Jet colormap.

The baseline CNN features [Fig. 7(b), (c), (g), and (h)] exhibit weak responses in occluded areas, with blurred edges and poor target–background discrimination—especially in Img-2 where the occluded truck blends into the rocky background. In contrast, MS-DSAM-enhanced features [Fig. 7(d), (e), (i), and (j)] consistently recover critical structures. The roof edge of the sedan and the wheel contour of the pickup

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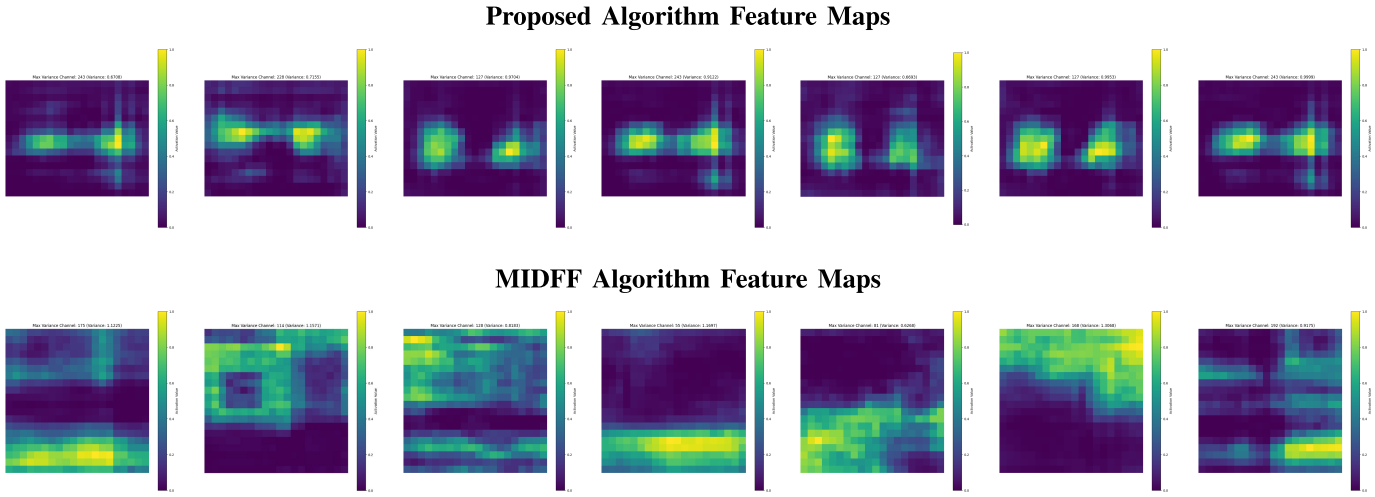


Fig. 6. Visualization of high-level semantic information extracted by the proposed algorithm (top row) and the CNN-LSTM algorithm (bottom row) for selected images of three categories (model cars, SUVs, and pickup trucks), highlighting differences in feature coherence and discriminability under occlusion.

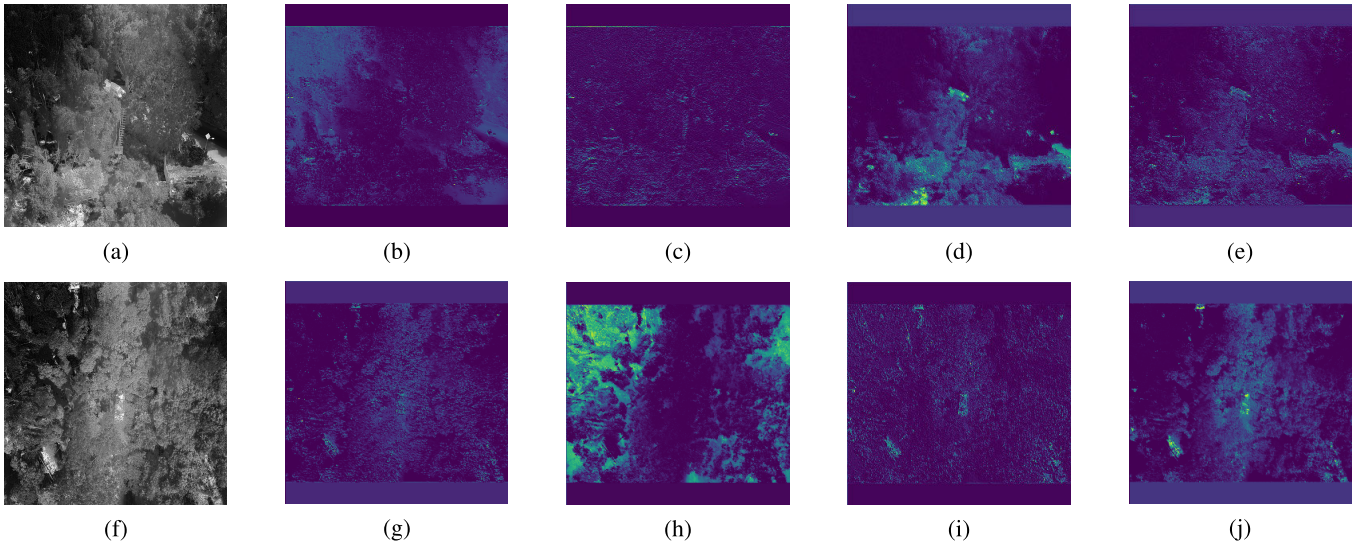


Fig. 7. Feature response visualization under occlusion. (Top row) Img-1 (partial vegetation occlusion). (Bottom row) Img-2 (heavy composite occlusion). (a) and (f) Original infrared images. (b), (c), (g), and (h) CNN backbone features. (d), (e), (i), and (j) MS-DSAM-enhanced features. Red dashed boxes highlight occluded regions. Brighter colors indicate stronger activations.

truck remain discernible. This is attributed to the dual-domain design—SSA enforces long-range continuity, while frequency-domain attention enhances occlusion-invariant spectral patterns. The result is a more robust low-level representation that directly contributes to improved detection accuracy under occlusion.

V. Conclusion

This article has presented a robust solution to the persistent challenge of infrared object detection in airborne surveillance, specifically targeting scenarios characterized by sparse viewpoints, dynamic motion, and inherent sensor uncertainties. We argue that traditional multiview paradigms, which necessitate dense camera arrays and precise static calibration, are fundamentally ill-suited for agile UAV deployments. To bridge this gap, we proposed the dual-platform collaborative occlusion-resistant framework, a lightweight yet powerful architecture designed to maximize information gain from minimal, asynchronous observations.

The core contributions of this work are summarized as follows.

1) *Robust Alignment Under Uncertainty:* We introduced the HGSR scheme. By synergizing coarse physical projection with fine-grained semantic affine correction, HGSR effectively mitigates the nonlinear distortions caused by GPS/IMU noise. This addresses the theoretical limitation of rigid geometric projections, ensuring precise pixel-level alignment without reliance on high-precision external calibration.
2) *Occlusion-Invariant Feature Learning:* We designed the MS-DSAM. This module innovatively models target features in both the spatial strip domain (for structural continuity) and the frequency domain (for spectral consistency). This dual-domain approach compensates for the lack of texture in infrared imagery, significantly enhancing robustness against severe occlusion.
3) *Weakly Supervised Consistency:* We formulated the DLL, which enforces geometric and semantic consis-

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tency across views. This allows the network to learn collaborative representations without requiring dense, pixel-level correspondence supervision, greatly reducing annotation costs.

Extensive evaluations on standard benchmarks (Wildtrack and MultiviewX) and our newly constructed DualView-Occlusion dataset confirm the superiority of our approach. Most notably, our method achieves state-of-the-art performance using only two mobile views, outperforming existing baselines that rely on three or four fixed cameras. Furthermore, the framework maintains a real-time inference speed of 23.31 FPS with only 3.62 M parameters, validating its practicality for resource-constrained edge computing on UAVs.

Future work will focus on extending this framework to heterogeneous sensor networks (e.g., infrared–visible fusion) to exploit complementary modalities and further optimizing the registration module for GPS-denied environments, paving the way for fully autonomous swarm intelligence.

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